

Temporal Integrity in Neuromorphic Systems — Experimental Pipeline

ANN Training → SNN Conversion → Dual-Domain Attack Evaluation → ATTF Defense → Manuscript

Dataset: Fashion-MNIST

60,000 training | 10,000 test | 28×28 greyscale | 10 object classes
Normalize: $x_{\text{norm}} = (x - 0.2860) / 0.3530$ (channel-wise z-score)
Calibration subset: 512 images sampled uniformly at random (seed=42)

Phase 1 — Train Baseline ANN (models/ann_model.py)

Architecture: Conv2d(1→32) → Conv2d(32→64) → Conv2d(64→128) → FC(1152→256) → FC(256→10)
All hidden activations: ReLU6 [0, 6] (no BatchNorm — preserves threshold calibration fidelity)
Classifier head: Dropout(p=0.4) before final linear layer
Optimizer: Adam lr=1e-3 weight_decay=1e-4
LR scheduler: StepLR step_size=7 gamma=0.5
Training: 20 epochs | batch_size=128 | CrossEntropyLoss | ~0.5M parameters

Result: 93.20% Top-1 Accuracy | 15.50 MFLOPs per inference

checkpoints/baseline_ann_best.pt

Phase 2a — Threshold Calibration (Rueckauer et al., Front. Neurosci. 2017)

Forward pass 512 calibration images through the trained ANN
Record 99.9th-percentile post-ReLU6 activation per layer: $\theta_l = P_{99.9} \{ a_l(x) : x \in D_{\text{calib}} \}$
Calibrated thresholds: $\theta_1 = 1.829$ $\theta_2 = 2.028$ $\theta_3 = 1.592$ $\theta_4 = 1.754$
No gradient updates — strictly a measurement pass

Phase 2b — Weight Transfer to snnTorch IF Neuron Network (models/snn_model.py)

Copy conv1,2,3 + fc1,fc2 weights verbatim — zero retraining required
Replace ReLU6 activations with Non-leaky Integrate-and-Fire (IF) neurons
IF dynamics (soft reset): $U(t) = U(t-1) + I(t) - S(t-1) \cdot \theta_l$
 $S(t) = \Theta(U(t) - \theta_l)$ [residual charge preserved above θ]
Observed spike sparsity per layer: Conv1 96.8% | Conv2 46.5% | Conv3 13.7% | FC1 0.5%

Simulate $T \in \{16, 32, 64, 128\}$ time steps

T=64 selected

Pareto Optimum T=64 → 90.59% accuracy | 4.18× energy saving over ANN

Checkpoint written: checkpoints/converted_snn.pt (referenced by all Phase 3 & 4 evaluations)

T-step Sweep Results

T=16	84.16%	13.72 MSynOps	0.885×
T=32	88.68%	30.66 MSynOps	1.978×
T=64 ★	90.59%	64.75 MSynOps	4.178×
T=128	90.88%	133.18 MSynOps	8.592×

checkpoints/converted_snn.pt

Spatial Domain

Temporal Domain

Phase 3a — Spatial FGSM Attack (phase3_adversarial.py)

Surrogate proxy: Use ANN (shared weights → full gradient access)
Compute gradient: $g = \nabla_x L(\text{CE}(\text{ANN}(x), y))$
Craft perturbation: $x_{\text{adv}} = \text{clamp}(x + \epsilon \cdot \text{sign}(g), x_{\text{min}}, x_{\text{max}})$
Evaluate on SNN: Feed x_{adv} to converted SNN (T=64, direct coding)
Sweep ϵ budgets: $\epsilon \in \{0.00, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$

FGSM Results (full 10,000-image test set)

$\epsilon=0.00$	90.59%	0.00 pp
$\epsilon=0.10$	61.04%	−29.55 pp
$\epsilon=0.20$	40.30%	−50.29 pp
$\epsilon=0.30$	22.70%	−67.89 pp

Saved: results/fgsm_accuracies.csv

$\epsilon=0.30 \rightarrow 22.70\%$ (−67.89 pp) CATASTROPHIC COLLAPSE

Shared weight inheritance = full surrogate vulnerability | Transfer attack succeeds completely

Phase 3b — Temporal Jitter Attack (phase3_adversarial.py)

Threat model: AER interconnect: clock skew + thermal drift + impedance noise
Per-neuron delay: $d \sim \mathcal{N}(0, \sigma^2)$ (half-normal, independent per (b,c,h,w) channel)
Clamp delay: $d_c = \min(\text{round}(d), T-1)$
Mask input current: $I(t; d_c) = x_{\text{norm}}$ if $t \geq d_c$, else 0
Sweep σ budgets: $\sigma \in \{0, 1, 2, 4, 6, 8, 10\}$ ms | $\eta = 1 - \sigma/(2m) / T$

Jitter Results (full 10,000-image test set)

$\sigma=0$ ms	90.59%	0.00 pp
$\sigma=2$ ms	89.79%	−0.80 pp
$\sigma=6$ ms	88.79%	−1.80 pp
$\sigma=10$ ms	85.10%	−5.49 pp

Saved: results/jitter_accuracies.csv

$\sigma=10\text{ms} \rightarrow 85.10\%$ (−5.49 pp) | 12× LESS VULNERABLE THAN FGSM

T=64 simulation horizon acts as natural low-pass charge-integration filter against timing noise

Phase 3c — Per-Class Jitter Analysis (phase3b_perclass_jitter.py)

Disaggregate accuracy across all 10 Fashion-MNIST object categories
Pullover: −21.80 pp (HIGH RISK) | Sandal: −13.20 pp | T-shirt/top: −11.50 pp
Bag: −0.20 pp (RESILIENT) | Ankle boot: +0.20 pp | Trouser: +1.50 pp
Shirt anomaly: +8.20 pp [jitter smooths boundary noise → latent realignment]

results/perclass_jitter.csv

Phase 4 — ATTF Defense (models/attf_snn.py | phase4_attf_defence.py)

Load converted_snn.pt weights → replace snnTorch IF neurons with ATTFNeuron instances
ATTFNeuron per-step: ① mem += current ② $\theta_{\text{eff}} = \theta_{\text{base}} \times S(t)$ ③ $\text{spk} = (\text{mem} \geq \theta_{\text{eff}})$
④ mem -= $\theta_{\text{eff}} \cdot \text{spk}$ [soft reset] ⑤ update ring buffer (len W)
⑥ $B(t) = \sum \text{history}[0..W-1]$
⑦ $B \geq K \rightarrow S = \min(S/y, 3.0)$ else $S = \max(S \cdot y, 1.0)$ [$y=0.9$]
Hyperparameter grid: $W \in \{3, 5, 8, 10\} \times K \in \{2, 3, 4, 5\} = 16$ configurations
Optimal: W=3, K=4 (shortest detection window + moderate burst threshold)

Defended (W=3,K=4) at $\sigma=10\text{ms}$: 85.57% (+0.47 pp) | Clean: 90.58% | FGSM: no effect (confirmed)

results/phase4_hparam_defended.npy (4×4 accuracy grid)

results/phase4_attf_w3_k4_measured.csv

Generate Figures & Compile Nature-Format Manuscript

generate_nature_figures.py → 6 publication figures at 300 DPI
Fig 1: Accuracy-efficiency Pareto | Fig 2: FGSM degradation curve
Fig 3: Temporal jitter robustness | Fig 4: Per-class vulnerability bar chart
Fig 5: ATTF defense comparison | Fig 6: Hyperparameter sensitivity heatmap
build_nature_manuscript.py → Nature-format DOCX | 16 references | 95 clickable internal citations

figures/ (figure1...figure6.png @ 300 DPI)

Temporal_Integrity_Neuromorphic_Systems_Nature_Format.docx

Core Finding: 12× Spatial-Temporal Vulnerability Asymmetry

FGSM ($\epsilon=0.30$): −67.89 pp vs. Temporal Jitter ($\sigma=10\text{ms}$): −5.49 pp

Spatial gradient attacks exploit shared weight geometry | Temporal charge integration provides intrinsic low-pass filtering against timing noise

Limitations & Future Directions

- Evaluated on Fashion-MNIST with feedforward CNN only — deeper residual SNNs and DVS event-camera datasets remain open
- Timing jitter modeled synthetically (half-normal distribution) — physical silicon characterization on Intel Loihi 2 / BrainScaleS needed
- Uniform ATTF hyperparameters applied across all layers — layer-specific or class-conditional tuning may improve recovery
- Future: composite dual-domain defense (spatial denoising + temporal ATTF) | Extend to LIF and recurrent SNN topologies